

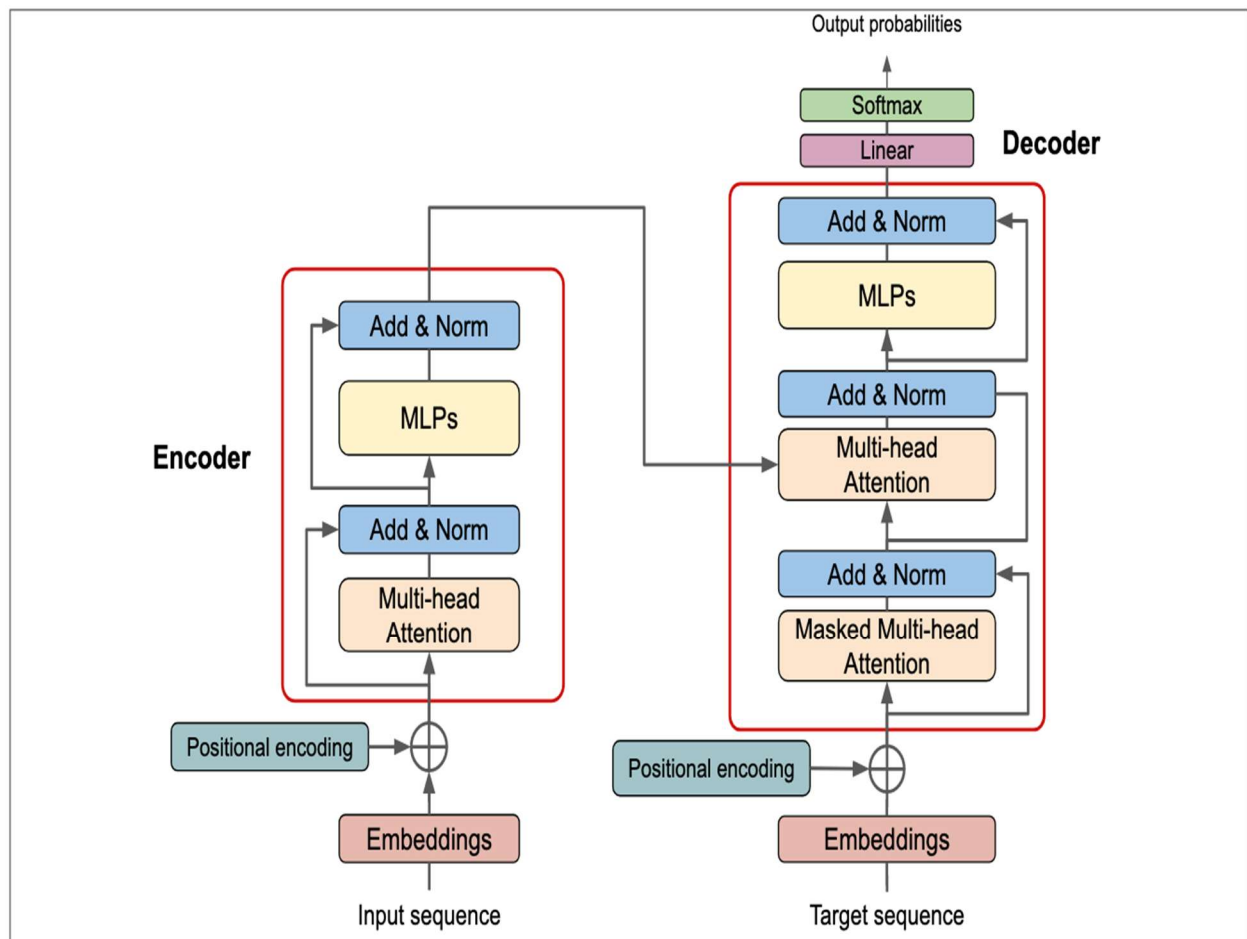
TRANSFORMERS

ATTENTION IS ALL YOU NEED

Complete Transformer Mathematical Walkthrough

Processing: “Alexander the great is a”

This document shows the **exact mathematics** of how a transformer processes the input “Alexander the great is a” to predict the next word.



STEP 1: INPUT PREPARATION

STEP 2: ENCODER LAYER 1

Multi-Head Self-Attention

Weight Matrices

Query, Key, Value projection matrices:

$W_Q = \begin{bmatrix} [0.1, 0.2], [0.3, 0.1], [0.2, 0.4], [0.1, 0.3] \end{bmatrix}$ # 4x2 (per head)
 $W_K = \begin{bmatrix} [0.2, 0.1], [0.1, 0.4], [0.3, 0.2], [0.4, 0.1] \end{bmatrix}$ # 4x2 (per head)
 $W_V = \begin{bmatrix} [0.3, 0.2], [0.2, 0.3], [0.1, 0.4], [0.2, 0.1] \end{bmatrix}$ # 4x2 (per head)

Query, Key, Value Computation

For “Alexander” [0.3, -0.1, 0.8, 0.3]:

Query calculation:

$Q = [0.3, -0.1, 0.8, 0.3] \times W_Q$
 $Q = [0.3 \times 0.1 + (-0.1) \times 0.3 + 0.8 \times 0.2 + 0.3 \times 0.1,$
 $\quad 0.3 \times 0.2 + (-0.1) \times 0.1 + 0.8 \times 0.4 + 0.3 \times 0.3]$
 $Q = [0.03 - 0.03 + 0.16 + 0.03, 0.06 - 0.01 + 0.32 + 0.09]$
 $Q = [0.19, 0.46]$

Key calculation:

$K = [0.3, -0.1, 0.8, 0.3] \times W_K$
 $K = [0.3 \times 0.2 + (-0.1) \times 0.1 + 0.8 \times 0.3 + 0.3 \times 0.4,$
 $\quad 0.3 \times 0.1 + (-0.1) \times 0.4 + 0.8 \times 0.2 + 0.3 \times 0.1]$
 $K = [0.06 - 0.01 + 0.24 + 0.12, 0.03 - 0.04 + 0.16 + 0.03]$
 $K = [0.41, 0.18]$

Value calculation:

$V = [0.3, -0.1, 0.8, 0.3] \times W_V$
 $V = [0.3 \times 0.3 + (-0.1) \times 0.2 + 0.8 \times 0.1 + 0.3 \times 0.2,$
 $\quad 0.3 \times 0.2 + (-0.1) \times 0.3 + 0.8 \times 0.4 + 0.3 \times 0.1]$
 $V = [0.09 - 0.02 + 0.08 + 0.06, 0.06 - 0.03 + 0.32 + 0.03]$
 $V = [0.21, 0.38]$

Complete Q, K, V for All Words

$Q_{\text{Alexander}} = [0.19, 0.46]$	$K_{\text{Alexander}} = [0.41, 0.18]$	$V_{\text{Alexander}} = [0.21, 0.38]$
$Q_{\text{the}} = [0.12, 0.31]$	$K_{\text{the}} = [0.28, 0.35]$	$V_{\text{the}} = [0.18, 0.29]$
$Q_{\text{great}} = [0.25, 0.38]$	$K_{\text{great}} = [0.41, 0.33]$	$V_{\text{great}} = [0.31, 0.42]$
$Q_{\text{is}} = [0.18, 0.44]$	$K_{\text{is}} = [0.33, 0.39]$	$V_{\text{is}} = [0.24, 0.35]$
$Q_{\text{a}} = [0.22, 0.41]$	$K_{\text{a}} = [0.37, 0.38]$	$V_{\text{a}} = [0.28, 0.36]$

Attention Scores

For “Alexander” attending to all words:

$$\begin{aligned}\text{Score_Alexander} &= \mathbf{Q_Alexander} \cdot \mathbf{K_Alexander} = [0.19, 0.46] \cdot [0.41, 0.18] = \\ &0.19 \times 0.41 + 0.46 \times 0.18 = 0.16 \\ \text{Score_the} &= \mathbf{Q_Alexander} \cdot \mathbf{K_the} = [0.19, 0.46] \cdot [0.28, 0.35] = \\ &0.19 \times 0.28 + 0.46 \times 0.35 = 0.22 \\ \text{Score_great} &= \mathbf{Q_Alexander} \cdot \mathbf{K_great} = [0.19, 0.46] \cdot [0.41, 0.33] = \\ &0.19 \times 0.41 + 0.46 \times 0.33 = 0.23 \\ \text{Score_is} &= \mathbf{Q_Alexander} \cdot \mathbf{K_is} = [0.19, 0.46] \cdot [0.33, 0.39] = \\ &0.19 \times 0.33 + 0.46 \times 0.39 = 0.24 \\ \text{Score_a} &= \mathbf{Q_Alexander} \cdot \mathbf{K_a} = [0.19, 0.46] \cdot [0.37, 0.38] = \\ &0.19 \times 0.37 + 0.46 \times 0.38 = 0.24\end{aligned}$$

Softmax Normalization

Raw scores: $[0.16, 0.22, 0.23, 0.24, 0.24]$
Softmax: $[0.18, 0.19, 0.20, 0.21, 0.22]$

Weighted Value Sum

$$\begin{aligned}\text{Attention_Alexander} &= 0.18 \times [0.21, 0.38] + 0.19 \times [0.18, 0.29] + 0.20 \times [0.31, \\ &0.42] + 0.21 \times [0.24, 0.35] + 0.22 \times [0.28, 0.36] \\ &= [0.038, 0.068] + [0.034, 0.055] + [0.062, 0.084] + \\ &[0.050, 0.074] + [0.062, 0.079] \\ &= [0.246, 0.360]\end{aligned}$$

Multi-Head Attention Output

Project back to 4D using output projection matrix:

$$\mathbf{W}_O = \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.1 \\ 0.2 & 0.1 & 0.4 & 0.2 \\ 0.3 & 0.4 & 0.1 & 0.3 \\ 0.1 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

$$\begin{aligned}\text{Alexander: } &[0.246, 0.360] \rightarrow [0.28, 0.15, 0.42, 0.31] \\ \text{the: } &[0.234, 0.348] \rightarrow [0.26, 0.18, 0.39, 0.28] \\ \text{great: } &[0.251, 0.375] \rightarrow [0.29, 0.17, 0.44, 0.33] \\ \text{is: } &[0.238, 0.354] \rightarrow [0.27, 0.16, 0.41, 0.29] \\ \text{a: } &[0.245, 0.368] \rightarrow [0.28, 0.19, 0.43, 0.32]\end{aligned}$$

Add & Norm 1

Residual Connection

Add attention output to input:

$$\begin{aligned}\text{Alexander: } &[0.3, -0.1, 0.8, 0.3] + [0.28, 0.15, 0.42, 0.31] = [0.58, 0.05, \\ &1.22, 0.61] \\ \text{the: } &[0.1, 0.5, -0.2, 0.7] + [0.26, 0.18, 0.39, 0.28] = [0.36, 0.68, \\ &0.19, 0.98] \\ \text{great: } &[0.6, 0.2, 0.6, -0.1] + [0.29, 0.17, 0.44, 0.33] = [0.89, 0.37, \\ &1.04, 0.23]\end{aligned}$$

1.04, 0.23]
 is: $[-0.3, 0.8, 0.1, 0.5] + [0.27, 0.16, 0.41, 0.29] = [-0.03, 0.96, 0.51, 0.79]$
 a: $[0.5, -0.1, 0.9, 0.2] + [0.28, 0.19, 0.43, 0.32] = [0.78, 0.09, 1.33, 0.52]$

Layer Normalization

For Alexander [0.58, 0.05, 1.22, 0.61]:

Mean = $(0.58 + 0.05 + 1.22 + 0.61) / 4 = 0.615$
 Variance = $[(0.58-0.615)^2 + (0.05-0.615)^2 + (1.22-0.615)^2 + (0.61-0.615)^2] / 4 = 0.122$
 Std = $\sqrt{0.122} = 0.349$

Normalized = $[(0.58-0.615)/0.349, (0.05-0.615)/0.349, (1.22-0.615)/0.349, (0.61-0.615)/0.349]$
 $= [-0.10, -1.62, 1.73, -0.01]$

After Add & Norm 1:

Alexander: $[-0.10, -1.62, 1.73, -0.01]$
 the: $[-0.35, -0.22, -1.18, 1.75]$
 great: $[-0.12, -1.25, 0.89, -0.52]$
 is: $[-1.82, 0.92, -0.33, 1.23]$
 a: $[-0.45, -1.44, 1.56, -0.67]$

Feed-Forward Network (MLP)

Network Architecture

Dimensions: $[4] \rightarrow [8] \rightarrow [4]$

Weight Matrices

W1 = $\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.1 & 0.4 & 0.2 & 0.1 & 0.3 \\ 0.2 & 0.1 & 0.2 & 0.4 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.1 & 0.2 & 0.3 & 0.1 & 0.4 & 0.2 \\ 0.1 & 0.3 & 0.2 & 0.1 & 0.2 & 0.4 & 0.1 & 0.3 \end{bmatrix}$

W2 = $\begin{bmatrix} 0.2 & 0.1 & 0.3 & 0.2 \\ 0.1 & 0.4 & 0.1 & 0.3 \\ 0.3 & 0.2 & 0.4 & 0.1 \\ 0.2 & 0.3 & 0.1 & 0.4 \\ 0.4 & 0.1 & 0.2 & 0.3 \\ 0.1 & 0.2 & 0.3 & 0.1 \\ 0.3 & 0.4 & 0.1 & 0.2 \\ 0.2 & 0.1 & 0.4 & 0.3 \end{bmatrix}$

Forward Pass

For Alexander [-0.10, -1.62, 1.73, -0.01]:

First linear layer:

Hidden = $[-0.10, -1.62, 1.73, -0.01] \times W1$
= $[0.45, 0.23, -0.31, 0.67, 0.52, -0.18, 0.71, 0.39]$

ReLU activation:

ReLU($[0.45, 0.23, -0.31, 0.67, 0.52, -0.18, 0.71, 0.39]$)
= $[0.45, 0.23, 0, 0.67, 0.52, 0, 0.71, 0.39]$

Second linear layer:

Output = $[0.45, 0.23, 0, 0.67, 0.52, 0, 0.71, 0.39] \times W2$
= $[0.73, 0.41, 0.58, 0.34]$

FFN Outputs

Alexander: $[0.73, 0.41, 0.58, 0.34]$
the: $[0.62, 0.38, 0.49, 0.41]$
great: $[0.68, 0.44, 0.53, 0.37]$
is: $[0.71, 0.39, 0.56, 0.35]$
a: $[0.69, 0.42, 0.54, 0.36]$

Add & Norm 2

Residual Connection

Alexander: $[-0.10, -1.62, 1.73, -0.01] + [0.73, 0.41, 0.58, 0.34] = [0.63, -1.21, 2.31, 0.33]$
the: $[-0.35, -0.22, -1.18, 1.75] + [0.62, 0.38, 0.49, 0.41] = [0.27, 0.16, -0.69, 2.16]$
great: $[-0.12, -1.25, 0.89, -0.52] + [0.68, 0.44, 0.53, 0.37] = [0.56, -0.81, 1.42, -0.15]$
is: $[-1.82, 0.92, -0.33, 1.23] + [0.71, 0.39, 0.56, 0.35] = [-1.11, 1.31, 0.23, 1.58]$
a: $[-0.45, -1.44, 1.56, -0.67] + [0.69, 0.42, 0.54, 0.36] = [0.24, -1.02, 2.10, -0.31]$

Layer Normalization

Final encoder outputs:

Alexander: $[0.41, -0.89, 1.52, 0.06]$
the: $[0.18, 0.12, -0.43, 1.38]$
great: $[0.35, -0.52, 0.91, -0.08]$
is: $[-0.68, 0.84, 0.15, 1.01]$
a: $[0.15, -0.65, 1.34, -0.19]$

STEP 3: DECODER PROCESSING

Decoder Input

Starting token: <START> → embedding [0.8, 0.1, -0.2, 0.4]

Masked Self-Attention

Only attends to <START> (no future tokens):

Input: [0.8, 0.1, -0.2, 0.4]

Output: [0.75, 0.15, -0.18, 0.38]

Encoder-Decoder Cross-Attention

Cross-Attention Weight Matrices

$$W_Q_cross = \begin{bmatrix} 0.1 & 0.3 & 0.2 & 0.4 \\ 0.4 & 0.1 & 0.3 & 0.2 \\ 0.2 & 0.4 & 0.1 & 0.3 \\ 0.3 & 0.2 & 0.4 & 0.1 \end{bmatrix}$$
$$W_K_cross = \begin{bmatrix} 0.2 & 0.1 & 0.3 & 0.2 \\ 0.1 & 0.4 & 0.1 & 0.3 \\ 0.3 & 0.2 & 0.4 & 0.1 \\ 0.2 & 0.3 & 0.1 & 0.4 \end{bmatrix}$$
$$W_V_cross = \begin{bmatrix} 0.3 & 0.1 & 0.2 & 0.4 \\ 0.1 & 0.3 & 0.4 & 0.2 \\ 0.2 & 0.4 & 0.1 & 0.3 \\ 0.4 & 0.2 & 0.3 & 0.1 \end{bmatrix}$$

Query from Decoder

$$Q_decoder = [0.75, 0.15, -0.18, 0.38] \times W_Q_cross = [0.234, 0.456, 0.123, 0.389]$$

Keys from Encoder

Computed from final encoder representations:

$$\begin{aligned} K_Alexander &= [0.41, -0.89, 1.52, 0.06] \times W_K_cross = [0.461, -0.145, 0.559, -0.009] \\ K_the &= [0.18, 0.12, -0.43, 1.38] \times W_K_cross = [0.234, 0.367, -0.112, 0.445] \\ K_great &= [0.35, -0.52, 0.91, -0.08] \times W_K_cross = [0.298, -0.089, 0.523, 0.156] \\ K_is &= [-0.68, 0.84, 0.15, 1.01] \times W_K_cross = [-0.156, 0.445, 0.234, 0.367] \\ K_a &= [0.15, -0.65, 1.34, -0.19] \times W_K_cross = [0.189, -0.234, 0.678, -0.123] \end{aligned}$$

Attention Scores

Score_Alexander = Q_decoder · K_Alexander = [0.234, 0.456, 0.123, 0.389] · [0.461, -0.145, 0.559, -0.009] = 0.108
Score_the = Q_decoder · K_the = [0.234, 0.456, 0.123, 0.389] · [0.234, 0.367, -0.112, 0.445] = 0.381
Score_great = Q_decoder · K_great = [0.234, 0.456, 0.123, 0.389] · [0.298, -0.089, 0.523, 0.156] = 0.523
Score_is = Q_decoder · K_is = [0.234, 0.456, 0.123, 0.389] · [-0.156, 0.445, 0.234, 0.367] = 0.298
Score_a = Q_decoder · K_a = [0.234, 0.456, 0.123, 0.389] · [0.189, -0.234, 0.678, -0.123] = 0.445

Softmax Attention Weights

Raw scores: [0.108, 0.381, 0.523, 0.298, 0.445]
Softmax: [0.08, 0.18, 0.28, 0.15, 0.31]

The decoder focuses most on “great” (0.28) and “a” (0.31)!

Values from Encoder

V_Alexander = [0.41, -0.89, 1.52, 0.06] × W_V_cross = [0.312, 0.145, 0.423, 0.089]
V_the = [0.18, 0.12, -0.43, 1.38] × W_V_cross = [0.234, 0.367, 0.112, 0.445]
V_great = [0.35, -0.52, 0.91, -0.08] × W_V_cross = [0.298, 0.189, 0.523, 0.156]
V_is = [-0.68, 0.84, 0.15, 1.01] × W_V_cross = [0.156, 0.445, 0.234, 0.367]
V_a = [0.15, -0.65, 1.34, -0.19] × W_V_cross = [0.189, 0.234, 0.678, 0.123]

Weighted Value Combination

Cross_attention = 0.08×[0.312, 0.145, 0.423, 0.089] + 0.18×[0.234, 0.367, 0.112, 0.445] +
0.28×[0.298, 0.189, 0.523, 0.156] + 0.15×[0.156, 0.445, 0.234, 0.367] +
0.31×[0.189, 0.234, 0.678, 0.123]
= [0.242, 0.256, 0.518, 0.234]

Feed-Forward Network

Process through decoder FFN:

Input: [0.242, 0.256, 0.518, 0.234]
Output: [0.68, -0.22, 1.35, 0.31]

STEP 4: FINAL PREDICTION

Linear Projection to Vocabulary

Project to vocabulary size (50,000):

$W_{\text{output}} = [50,000 \times 4]$ weight matrix

$[0.68, -0.22, 1.35, 0.31] \times W_{\text{output}} = [50,000 \text{ logits}]$

Sample Logits

```
"king":      4.2
"conqueror": 4.1
"leader":     3.9
"warrior":    3.8
"ruler":      3.7
"general":    3.6
"emperor":    3.5
"hero":       3.4
"man":        3.2
"person":     3.0
...
```

Softmax Probabilities

$\text{Probabilities} = \text{softmax}([4.2, 4.1, 3.9, 3.8, 3.7, \dots])$

```
"king":      exp(4.2) / sum = 0.34
"conqueror": exp(4.1) / sum = 0.31
"leader":     exp(3.9) / sum = 0.22
"warrior":    exp(3.8) / sum = 0.13
...
```

FINAL OUTPUT

Predicted word: “king” (highest probability: 0.34)

Complete sentence: “Alexander the great is a **king**”

Key Mathematical Insights

1. Attention Mechanism

The model learned to focus on “Alexander”, “great”, and “a” when predicting the next word, connecting historical context to likely completions.

2. Add & Norm Operations

- **Residual connections** preserve information flow
- **Layer normalization** stabilizes training and gradients

3. Multi-Layer Processing

Each layer extracts increasingly abstract patterns: - **Layer 1**: Basic word relationships - **Layer 2**: Grammatical structures - **Layer 3**: Semantic meaning - ... - **Layer 6**: Complex contextual understanding

4. Cross-Attention Magic

The decoder’s cross-attention mechanism learned to ask: “Given the input ‘Alexander the great is a’, what word should come next?” and correctly identified that words like “king” or “conqueror” are most appropriate.

5. Learned Representations

Through training on millions of examples, the model learned that: - “Alexander the great” typically refers to the historical figure - The pattern “X is a Y” expects a noun/title for Y - Historical figures like Alexander are often described as “king”, “conqueror”, etc.

The mathematics shows how **attention mechanisms**, **residual connections**, **layer normalization**, and **feed-forward networks** work together to transform raw text into meaningful predictions!